Numerical Simulation of the 1-D Two-Fluid Four-Equation Model in Pipelines Using the Very High Order Flux Reconstruction (FR) Method

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ABSTRACT

Understanding multiphase flow in oil and gas pipelines is essential for addressing flow assurance challenges. One-dimensional (1-D) models are widely used in industrial applications due to their low computational cost; however, commercial simulators typically rely on low-order numerical schemes, which may compromise accuracy. In this work, we present, for the first time, a successful very high-order implementation of the Flux Reconstruction (FR) method applied to the 1-D two-fluid four-equation model. The proposed approach combines a high-order FR formulation with a hierarchical limiter and a staggered grid arrangement, ensuring stability and accuracy in the simulation of multiphase flows. Within the FR framework, we implement and analyze correction functions that recover formulations equivalent to the Nodal Discontinuous Galerkin (NDG) and Spectral Difference (SD) methods. Pressure–velocity coupling is handled using a semi-implicit SIMPLE scheme, with the pressure-correction equation solved via the Two-Point Flux Approximation (TPFA). The method is assessed using five benchmark problems from the literature and validated against results from the commercial simulator ALFAsim. The proposed approach achieves accuracy comparable to fine-mesh commercial simulations while using significantly fewer control volumes. These results demonstrate the potential of high-order FR methods for efficient and accurate flow assurance simulations in oil and gas pipelines.

1. Introduction

Simulating multiphase flow in pipelines is essential for effective process management across various engineering fields. Key parameters, including phase velocities, volumetric fractions, pressures, and densities, play a critical role in ensuring flow continuity (Su (2003)). In the oil industry, accurately predicting these parameters is particularly vital for managing flow assurance in oil pipelines.

Flow assurance involves addressing challenges such as scale formation, hydrate and wax deposition, pipeline erosion, and corrosion, all of which are critical to preventing significant human and financial losses (Su (2003)). Predicting these phenomena in pipelines, particularly in oil wells and risers that can have thousands of meters of length is highly complex. In this context, numerical simulation of fluid flow via Computational Fluid Dynamics (CFD) provides a viable approach for making these predictions, however, due to the high computational costs associated with the 3-D or even the 2-D numerical modeling of the multiphase flow in oil pipelines often, substantial simplifications are required. One of the most common simplifications is treating the flow as one-dimensional (1-D), which is widely adopted by commercial simulators designed for these analyses, such as ALFAsim (ESSS (2024)), LedaFlow (Danielson, Bansal, Djoric, Duret, Johansen and Hellan (2011)) and OLGA (Bendlksen, Malnes, Moe and Nuland (1991)).

When two phases are considered in the flow, the 1-D two-fluid model is the most common mathematical representation of the physics governing both phases (Jerez-Carrizales, Jaramillo and Fuentes (2015), van Zwieten,

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Sanderse, Hendrix, Vuik and Henkes (2017), Wang, Gong and Wu (2015), Ishii and Hibiki (2006), Štrubelj and Tiselj (2011), Holmås, Clamond and Langtangen (2008), Leporini, Bonzanini, Ferrari and Poesio (2021)). This model consists of a set of non-linear partial differential equations that describes mass, momentum, and energy conservation for each phase (Ishii and Hibiki (2006)). Various formulations of the one-dimensional two-fluid model are found in the literature, each one with different approaches to derive the discrete governing equations and their computational implementation (Jerez-Carrizales et al. (2015), van Zwieten et al. (2017), Wang et al. (2015), Ishii and Hibiki (2006), Prosperetti and Tryggvason (2007)).

A commonly referenced variant in literature is the two-fluid four-equation single-pressure model (Wang et al. (2015)), which assumes that both phases share the same pressure field. This model can be considered the isothermal case of the two-fluid six-equation single-pressure model (Jerez-Carrizales et al. (2015)). It is also employed in reactor thermal-hydraulic analysis codes, such as RELAP (Hassan and Morgan (1983)), TRAC (Liles and Mahaffy (1986)), and CATHARE (Barre and Bernard (1990), Pourgol-Mohammad (2009)), and in several studies involving multiphase flow in oil pipelines, such as Cazarez-Candia and Piedra-González (2017), Figueiredo, Baptista, Freitas Rachid and Bodstein (2017), Sondermann, Baptista, Bastos de Freitas Rachid and Bodstein (2019) , Figueiredo, Baptista, Rachid and Bodstein (2021) , M. H. Yahyanezhad, M. Abbasi and V. Shokri (2024) , Li, Tian, He, Lv, Zhou and Li (2024).

Two-fluid single-pressure models are relatively easy to implement, however, they are susceptible to instabilities, particularly the Kelvin-Helmholtz instability (De Bertodano, Fullmer, Clausse and Ransom (2016)), which occurs when two fluids are forced to flow together under a shared pressure field. This instability arises from the loss of hyperbolicity of the system of equations (De Bertodano et al. (2016), Bestion (1990) , Toumi and Raymond (2005)). One of the main strategies to overcome this problem is to introduce an artificial interfacial pressure difference (Wang et al. (2015), De Bertodano et al. (2016), Bestion (1990), Toumi and Raymond (2005)) to ensure the hyperbolicity of the system. Two primary approaches for this are the CATHARE model (Bestion (1990)) and the model proposed by Toumi (Toumi and Raymond (2005)). Both are quite similar and are generally used for different flow regimes. However, the "Toumi model", despite having a simpler formulation, is more dependent on its adjustable parameter.

Staggered grids and first order donor-cell differencing schemes are commonly used to achieve stable results in the two-fluid single-pressure model (van Zwieten et al. (2017), Toumi (1996)). However, these schemes may introduce excessive numerical diffusion, which reduces solution accuracy. To ensure higher precision, advanced high-order schemes must be employed. Notable examples include the Monotonic Upstream-centered Scheme for Conservation Laws (MUSCL) (van Leer (1979), Contreras, Lyra, Souza and Carvalho (2016), Contreras, Vaz, Pacheco, Antunes, Lyra and Carvalho (2023) , Contreras, Carvalho, Galindez-Ramirez and Lyra (2021)), the Spectral Volume (SV) method by Wang (2002), Wang and Liu (2004), Wang and Liu (2005), Wang and Liu (2006) , Galindez-Ramirez, Souza, Carvalho and Lyra (2017), the Spectral Difference (SD) method by Liu, Vinokur and Wang (2006), the Discontinuous Galerkin (DG) method by Reed and Hill (1973), Fernandez, Christophe, Terrana, Nguyen and Peraire (2018), Ferrer, Rubio, Ntoukas, Laskowski, Mariño, Colombo, Mateo-Gabín, Marbona, Manrique de Lara, Huergo, Manzanero, Rueda-Ramírez, Kopriva and Valero (2023), Antonietti, Mazzieri, Muhr, Nikolić and Wohlmuth (2020), and the Flux Reconstruction (FR) method by Huynh (2007). The latter has recently been adapted for multiphase and multicomponent fluid flow in porous media (Contreras et al. (2021), Galindez-Ramirez, Carvalho and Lyra (2020a), Galindez-Ramirez, Contreras, Carvalho and Lyra (2020b), Galindez-Ramirez, Contreras, Carvalho and Lyra (2022), Galindo, Lacerda, Galindez-Ramirez, Lyra and Carvalho (2023)). Furthermore, the FR serves as a unifying framework capable of recovering specific versions of the Nodal Discontinuous Galerkin (NDG), Spectral Volume (SV) and Spectral Difference (SD) methods, with a differential form, by employing different correction functions (Galindez-Ramirez et al. (2020a), Galindez-Ramirez et al. (2020b), Galindez-Ramirez et al. (2022) , Galindo et al. (2023)), allowing, not only, the recovering of such methods, but also of creating new ones. These functions are used to correct the discontinuous flux across each control surface within the computational domain.

In addition to unifying several other formulations, the Flux Reconstruction (FR) method presents several additional advantages. It features minimal inter-element connectivity, as only fluxes at the element interfaces need to be exchanged between neighboring cells. This property facilitates implementation, reduces communication overhead, and enhances parallel scalability. Compared to the DG method, FR requires fewer volumetric integration operations, making it computationally lighter. Moreover, the FR framework allows for the modular coupling of different Riemann solvers in a straightforward manner and is naturally compatible with high-order limiting techniques, such as the Multi-dimensional Limiting Process (MLP) and WENO-type reconstructions, enabling effective control of spurious oscillations near discontinuities. These features make the FR approach both computationally efficient and highly adaptable for simulating complex flow regimes (Huynh (2007), Wang and Gao (2009), Vincent, Castonguay and Jameson (2011)).

In literature, there are several implementations of 2nd and very high-order (>2) methods applied to the two-fluid model for 1-D two-phase flow. Among them, the study of van Zwieten et al. (2017) stands out for successfully implementing a space-time Discontinuous Galerkin (DG) method in the two-fluid six-equation model, also employing h-type mesh adaptation. In a different context, Dumbser, Castro, Parés and Toro (2009) introduced the ADER (Arbitrary High-Order DERivative) framework for solving, among other systems, a variant of the two-fluid four-equation single-pressure model. The ADER approach generalizes the Godunov method to arbitrary high orders of accuracy by employing a local space–time DG predictor combined with high-order flux evaluations, allowing the accurate resolution of discontinuities while minimizing numerical dissipation. Another noteworthy work is the one of Wu, Chao, Wu, Shan and Gou (2017), who implemented various TVD spatial discretization schemes in a MUSCL-like framework, achieving second-order accuracy in the two-fluid seven-equation model. In this paper, authors initially assumed that this model would be unconditionally stable, however, their results exhibited spurious oscillations. Munkejord (2005) investigated two types of two-fluid models: the four-equation single-pressure two-fluid model and the five-equation two-pressure model, the latter requiring a pressure relaxation procedure to enforce mechanical equilibrium between phases. In addition, Munkejord (2005) compared several numerical schemes, including Roe-type approximate Riemann solvers, flux-limiter centered (FLIC) schemes, and multi-stage centered (MUSTA) methods, emphasizing their accuracy, robustness, and computational efficiency across various benchmark problems. Sondermann et al. (2019), in turn, implemented the flux-corrected transport (FCT) method in the two-fluid five-equation model, focusing on horizontal flow with a stratified flow pattern in oil pipelines. They compared their results with those from the commercial simulator OLGA, demonstrating excellent agreement. More recently, Hu and Kozlowski (2022) implemented a new Weighted Essentially Non-Oscillatory (WENO) numerical formulation for the two-fluid six-equation model, achieving up to third-order accuracy for smooth solutions and obtaining stable results.

The present authors, in the conference paper (Do Nascimento, De Carvalho, Da Silva, Lyra and Antunes (2023)), have previously reported an initial attempt to implement the Flux Reconstruction (FR) method within the framework of the two-fluid four-equation model. In that preliminary study, a co-located mesh arrangement was adopted, and no appropriate limiting strategy was incorporated. Consequently, the formulation exhibited severe numerical instabilities, even for second-order polynomial reconstructions. As a result, only a single-phase smooth flow case could be successfully simulated. When the classical Discontinuity Movement in the Uniform Flow benchmark problem (Yeom and Chang (2011)) was considered, the solution displayed pronounced spurious oscillations within the first milliseconds of transient evolution.

In this context, the present study reports, for the first time in the literature, the successful development and implementation of a very high-order Flux Reconstruction (FR) method for the discretization of the two-fluid four-equation model. Specifically, the FR formulation is employed to solve the mass and momentum conservation equations of the two-fluid, four-equation, single-pressure model. We used a staggered grid approach and the interpolation of velocities and volume fractions between the meshes was unnecessary, as the reconstructions from the FR formulation were used directly. We also test two correction functions the one that leads to a version of Nodal Discontinuous Galerkin (NDG) method, and the other leading to a version of the Spectral Difference (SD) method. To evaluate the potential of our approach, we present and discuss numerical results for five benchmark problems found in the literature (Wang et al. (2015), Morales-Ruiz, Rigola, Rodriguez and Oliva (2012)).

The outline of this paper is as follows. In Section 2, we recall the governing equations of the compressible two-fluid four-equation single-pressure model, and we also outline the artificial pressure difference model used to stabilize the model, i.e., CATHARE model. In Section 3, we describe the application of the FR method to discretize the two-fluid four-equation model, and we also present the details of its implementation in the context of a staggered grid and the classical SIMPLE scheme which are used for the pressure-velocity coupling. In Section 4, we present the results of five representative test cases and, finally, in Section 5, we present the conclusions of our work.

2. Mathematical formulation

The two-fluid four-equation single-pressure model consists of two continuity equations and two momentum equations, adapted for 1-D, compressible and isothermal flow (Wang et al. (2015), Figueiredo et al. (2017), Figueiredo et al. (2021),Cortes, Debussche and Toumi (1998)). In this model, the pressure is assumed to be shared by both phases. Unlike the two-pressure two-fluid models, in the shared-pressure formulation the pressure relaxation process is unnecessary, as the instantaneous mechanical equilibrium between phases is already embedded in the model formulation (Munkejord (2005), Figueiredo et al. (2017)). This version of the model is adapted from the one used by Wang (Wang et al. (2015)).

The continuity equation for each phase k, with $k = 1, 2$ can be expressed as:

$$\frac{\partial(\alpha_k \rho_k)}{\partial t} + \frac{\partial(\alpha_k \rho_k u_k)}{\partial x} = 0 \tag{1}$$

where α_k represents the volumetric fraction occupied by phase k and ρ_k and u_k represent the density and the velocity of phase k, respectively. Following, we can write the momentum equation for phase k, as (Wang et al. (2015)):

$$\frac{\partial(\alpha_k \rho_k u_k)}{\partial t} + \frac{\partial\left(\alpha_k \rho_k u_k^2\right)}{\partial x} = -\alpha_k \frac{\partial p_k}{\partial x} - \left(p_k - p_k^i\right) \frac{\partial \alpha_k}{\partial x} + \alpha_k \rho_k g \sin(\theta) + \\ + \frac{4\tau_{k,W}}{D_{H,k}} + \frac{\alpha \tau_i S_i}{A_k} \tag{2}$$

where p_k is the pressure for phase k, p_k^i is the interface pressure for phase k (Cortes et al. (1998)), g is the local gravitational acceleration, θ is the pipe slope, $D_{H,k}$ is the hydraulic diameter, A_k is the cross-sectional area occupied by phase k, S_i is the interfacial perimeter, τ_i and $\tau_{k,W}$ are the interfacial and wall friction shear stresses, respectively (Cazarez-Candia and Piedra-González (2017)). In this case, the influence of mass transfer between the phases is ignored. To close the system of equations, we use the simple volumetric fraction restriction equation:

$$\alpha_1 + \alpha_2 = 1 \tag{3}$$

2.1. Pressure Correction term

Strictly speaking, since the pressure is shared between the two phases, the interfacial pressure difference term $\Delta p = p_k - p_k^i$ in Eq. 2 should be zero.

However, it is well known that some diffusion must be artificially introduced to stabilize the solution for general applications (Wang et al. (2015), Cortes et al. (1998)). However, due to the model's stability issues, an artificial, non-physical term, is introduced to add numerical diffusion. In literature, there are different approaches to approximate the value for the artificial interfacial pressure difference term. The most common approaches are the CATHARE (Bestion (1990)) and Toumi models (Toumi and Raymond (2005)). Wang et al. (2015) analyze the hyperbolicity of the two-fluid four-equation model and demonstrate the necessity of adopting a term such as the artificial interfacial pressure difference. In general, the two-fluid models, as well as other models in which the fluids are described as an intertwined continuum, yield an ill-posed mathematical problem (Ansari and Shokri (2007), Lhuillier, Chang and Theofanous (2013), Clausse and López De Bertodano (2021)). The Kelvin-Helmholtz (K-H) instability is the physical mechanism that causes this problem, which arises whenever two fluids are forced to flow together, sharing the pressure field and having different velocities.

In reference Toumi and Raymond (2005), authors propose that Δp must be defined in a consistent way to represent the physics of the problem, suggesting the following approximation for the pressure difference at the interface of the two phases:

$$\Delta p = \xi \rho_1 \rho_2 (u_2 - u_1)^2 \tag{4}$$

where ξ is an adjustable "shape parameter" that is used in an attempt to reinforce the hyperbolicity of the problem. In reference Toumi and Raymond (2005), authors also justified the choice of this pressure correction term because the pressure difference at the interface no longer exists if u_1 approaches u_2. In reference Omgba-Essama (2004), authors show that ξ has a lower limit for the simple pressure model in order to guarantee the hyperbolicity of the problem, which is given by:

$$\xi > \frac{\rho_2 \alpha_1}{\rho_2 \alpha_1 + \rho_1 \alpha_2} \tag{5}$$

In reference Pokharna, Mori and Ransom (1997), authors define this term as a "pressure coefficient". In practice, as $\xi = \frac{\rho_2 \alpha_1}{\rho_2 \alpha_1 + \rho_1 \alpha_2}$, authors in reference rewrite this term as being a lower limit value $\xi_0 = \frac{\rho_2 \alpha_1}{\rho_2 \alpha_1 + \rho_1 \alpha_2}$ multiplied by another

parameter γ, where $\gamma > 1$, i.e., $\xi = \gamma\xi_0$. The choice of the adjustable ξ term depends on the simulation variables. The reference Coquel, El Amine, Godlewski, Perthame and Rascle (1997) use $\xi = 1$ while in reference Omgba-Essama (2004) authors use different values of γ and graphically compare the relative velocities between the gas and the liquid together evaluating their influence over Kelvin–Helmholtz instabilities. As recommended by Evje and Flåtten (2003) and adopted by Wang et al. (2015), we have used $\gamma = 1.2$. On the other hand, in the so-called CATHARE model we formulate the artificial pressure difference term in a different way, which operates in a more straightforward manner, and less dependence on manual parameter adjustment Barre and Bernard (1990), Bestion (1990) . In this context, in reference Bestion (1990) authors propose that Δp is given by:

$$\Delta p = \gamma \frac{\rho_2 \alpha_1 \rho_1 \alpha_2}{\rho_2 \alpha_1 + \rho_1 \alpha_2} (u_2 - u_1)^2 \tag{6}$$

In the specific case of equal phase velocities, the physical mechanism for K-H instability is absent. Consequently, the vanishing of the Δp term under this condition is not a numerical issue, but rather the physically consistent and correct behavior of the model.

2.2. Thermodynamic Submodels

To account for the compressibility of the phases, considering phase k, as in Wang et al. (2015) and Morales-Ruiz et al. (2012), we assume the following simplified linear thermodynamic relations:

$$p_k = c_k^2 \left(\rho_k - \rho_k^0\right)^2 \tag{7}$$

The density change rate is considered constant and given by:

$$\Delta \rho_k = \frac{1}{c_k^2} \Delta p_k \tag{8}$$

where c_k is the speed of sound in phase k.

2.3. Initial and Boundary Conditions

To solve the system of equations, Eqs. (1-2), we need a proper set of initial and boundary conditions, which are given below. The initial pressure is given by:

$$p^0(x) = p(x, t = 0), \forall x \in \Omega, \tag{9}$$

the initial velocities

$$u_k^0(x) = u_k(x, t = 0), \forall x \in \Omega \tag{10}$$

and the initial volumetric fractions

$$\alpha_k^0(x) = \alpha_k(x, t = 0), \forall x \in \Omega \tag{11}$$

The boundary conditions are defined using Dirichlet or Neumann conditions:

$$u_k(x, t) = G_D \ \ on \ \ \Gamma_D \times [0, t] \tag{12}$$

$$\alpha_k(x, t) = H_D \ \ on \ \ \Gamma_D \times [0, t] \tag{13}$$

$$\frac{\partial p}{\partial x}(x, t) = G_N \ \ on \ \ \Gamma_N \times [0, t] \tag{14}$$

Where, in Eqs. (12-14) G_D, H_D and G_N represents, respectively, the prescribed velocities and volume fractions for phase k on the Dirichlet boundary, Γ_D and the prescribed pressure gradient on the Neumann boundary, Γ_N.

3. Numerical Formulation

In this section, we present the numerical strategy used to approximate the PDEs used in the two-fluid four-equation single-pressure model.

3.1. Discretization

The governing equations were integrated with respect to the local average fluid variables using the finite volume method (Patankar (1980)). The discretization employs the classical staggered grid arrangement (figure 1), with a primal (solid lines) and a dual grid (dashed line), where scalar variables (e.g., pressure, density, and volumetric fractions) are defined at the control volumes of the primal grid, while vector variables (e.g., velocities and fluxes) are defined at the control surfaces of the grid (Morales-Ruiz et al. (2012), Patankar (1980), Maliska (2023)). This grid configuration is adopted to handle the odd-even decoupling problem in pressure-correction algorithms (Patankar (1980)). A schematic representation of excerpt of the grid used and the localization of some variables is provided in Figure 1. The letters w, e and ee denote neighboring control surfaces, whereas W, P, E and F represent the control volumes of the considered excerpt of the primal grid, W^S, P^S, E^S and F^S correspond to the control volumes of the dual grid.

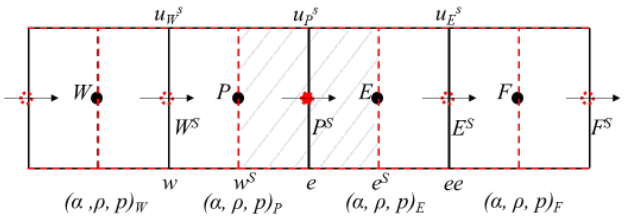


Figure 1: Control volumes and the staggered grid arrangement (adapted from Morales-Ruiz et al. (2012)).

Before discretizing the equations, the momentum equation, Eq. 2, is manipulated to linearize the advective term and thus facilitate its numerical solution (Wang et al. (2015)). First, the phase k continuity equation, Eq. 1, is multiplied by the phase k velocity (u_k), and the resulting expression is then subtracted from the phase k momentum equation, yielding the following equation:

$$\left(\alpha_k \rho_k\right) \frac{\partial u_k}{\partial t} + (\alpha_k \rho_k u_k) \frac{\partial u_k}{\partial x} = -\alpha_k \frac{\partial p_k}{\partial x} - \Delta p \frac{\partial \alpha_k}{\partial x} + \alpha_k \rho_k g \sin\theta + \tag{15}$$

$$+ \frac{4\tau_{k,W}}{D_{H,k}} + \frac{\alpha_k \tau_i S_i}{A_k}$$

By discretizing the Eq. 15 in the dual grid by the finite-volume method, the left-hand side (LHS) is written as:

$$LHS = \left(\alpha_k \rho_k\right)_e^n \frac{u_{k,e}^{n+1} - u_{k,e}^n}{\Delta t} + (\alpha_k \rho_k u_k)_e^n \left(\frac{u_k}{\Delta x}\right)_e^n \tag{16}$$

where the superscript n refers to the time level. The advective term $\frac{u_k}{\Delta x}$ are evaluated as a function of the numerical scheme. In the present work, we test two classical methods - the First Order Upwind (FOU) method and the Monotone Upstream-centered Scheme for Conservation Laws (MUSCL) - and the FR method, which is detailed in the following sections. The temporal discretization is performed explicitly, more details are also given later. The scalar variables α_k and ρ_k are evaluated on the dual grid by interpolation. In classical schemes such as the FOU method and the MUSCL, this interpolation is typically performed using an upwind approach (Wang et al. (2015), Maliska (2023)), as also adopted in this work. However, in the FR method this extra interpolation is obviously unnecessary because variables are reconstructed intrinsically during the reconstruction step of the formulation. The right-hand side (RHS) discretization of Eq. 15 follows. The advective terms of this part of the equation are discretized using the Control Volume Finite Difference Method (CVFD):

$$RHS = -\alpha_{k,e}^{n}\frac{p_E^{n+1}-p_P^{n+1}}{\Delta x} - \Delta p_{k,e}^{n}\frac{\alpha_{k,E}^{n}-\alpha_{k,P}^{n}}{\Delta x} + \left(\alpha_k\rho_k\right)_e^n g\sin\theta + \\ + \left(\frac{4\tau_{k,W}}{D_{H,k}}\right)_e^n + \left(\frac{\alpha_k\tau_i S_i}{A_k}\right)_e^n \tag{17}$$

The mass conservation equations, Eq. 1, are discretized using the primal grid. As in the previous equation, the temporal discretization is done explicitly, as follows:

$$\frac{(\alpha_k\rho_k)_P^{n+1}-(\alpha_k\rho_k)_P^{n}}{\Delta t} + \left(\frac{\alpha_k\rho_k u_k}{\Delta x}\right)_P^n = 0 \tag{18}$$

3.2. Flux Reconstruction Method

In the Flux Reconstruction (FR) method, the advective flux is continuously reconstructed across the domain using interpolation with Lagrange polynomials (Contreras et al. (2023), Huynh (2007), Galindez-Ramirez et al. (2022), Galindo et al. (2023), Romero, Asthana and Jameson (2016)). Initially, the unknown variable is interpolated and recovered in a discontinuous manner. Subsequently, the discontinuous polynomial representation of the flux is derived. Finally, the continuous flux is obtained by applying a correction to the discontinuous flux using an approximate Riemann solver (Huynh (2007)). The points used to construct the polynomial for the unknown variable are termed solution points, and their placement is determined by a selected quadrature.

The points used for flux reconstruction, which include both the solution points and the interface points, are referred to as flux points. To introduce a simplified formulation, a review of the existing FR method for the advective term is necessary. Consider a one-dimensional scalar conservation law, where the same generic advected variable, ϕ, is used to represent the advected quantities in this work, where ϕ denotes either α_k, ρ_k or u_k .

$$\frac{\partial\phi}{\partial t} + \frac{\partial f(\phi)}{\partial x} = 0 \tag{19}$$

where x is the spatial coordinate, t is time, $f = f(\phi)$, and is the advective flux, defined on a closed domain Ω . In the present work, for the continuity equations $f(\alpha_k u_k) = \alpha_k\rho_k u_k$, while for the momentum equations $f(u_k) = u_k$. Consider partitioning Ω into N control volumes, each denoted by Ω_i. The solution ϕ is approximated by numerical solutions ϕ^D which are defined as polynomials of degree P within each Ω_i. The piecewise sum of the polynomials provide a discontinuous solution ϕ^D throughout the Ω domain. Similarly, the flux $f(\phi)$is approximated by fluxes f_i^D which are defined as polynomials of degree $P+1$ within each Ω_i . The piecewise sum of these polynomials results in the discontinuous flux f^D throughout the Ω domain:

$$\phi^D = \sum_{i=0}^{N-1}\phi_i^D \approx \phi, \quad f^D = \sum_{i=0}^{N-1} f_i^D \approx f \tag{20}$$

Each subdomain Ω_i is mapped using an isoparametric transformation, which converts it into a standard reference control volume $\Omega_{is} = \{\, s \mid -1 < s < 1\}$ (Huynh (2007)). This parameterization enables the use of uniform polynomial bases across all control volumes.

$$s = 2\left(\frac{x-x_i}{x_{i+1}-x_i}\right) - 1 \tag{21}$$

Applying this transformation, ϕ_i^D can be rewritten as $\hat{\phi}_i^D$ and f_i^D as $\hat{f}_i^D$ (Huynh (2007)). The transformed equation within the $i-th$ standard control volume Ω_{is} has the following form:

$$\frac{\partial\hat{\phi}_i^D}{\partial t} + \frac{1}{J_i}\frac{\partial\hat{f}_i^D}{\partial s} = 0 \tag{22}$$

J_i is the Jacobian of the transformation, $J_i = \dfrac{x_{i+1} - x_i}{2}$. In one space dimension, on the reference control volume, the Lagrange polynomials can be written, using the solution points, as:

$$L_j(s) = \prod_{l=1, l\neq j}^{p+1} \frac{s - s_l}{s_j - s_l}, \quad j = 1, 2, \cdots, P+1 \tag{23}$$

where s_l are the point coordinates, with $l = 1, ..., P+1$, as depicted in Figure 2.

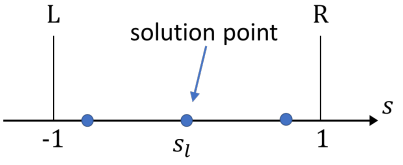


Figure 2: Solution point on reference domain, including cell interfaces.

Equation 22 is approximated within each control volume by representing the discontinuous solution $\hat{\phi}_i^D$ and the corresponding discontinuous flux $\hat{f}_i^D$ as expansions using the Lagrange polynomials defined on a set of $P+1$ interior solution and flux points:

$$\hat{\phi}_i^D = \sum_{j=1}^{P+1} \hat{\phi}_{i,j}^D L_j \tag{24}$$

$$\hat{f}_i^D = \sum_{j=1}^{P+1} \hat{f}_{i,j}^D L_j \tag{25}$$

where $\hat{\phi}_{i,j}^D$ and $\hat{f}_{i,j}^D$ are the known solution and flux values, respectively, at the $j-th$ solution/flux point in the i-th control volume, after parameterization, Eq. 21, and L_j are the corresponding Lagrange polynomials, Eq. 23. It is observed that the discontinuous solution and flux are both polynomials of degree P.

The continuous flux function $\hat{f}_i^C$ is constructed as:

$$\hat{f}_i^C = \hat{f}_i^D + \left(f_{i-1/2} - \hat{f}_{i,\,i-1/2}^D\right) g_L + \left(f_{i+1/2} - \hat{f}_{i,\,i+1/2}^D\right) g_R \tag{26}$$

where the terms $\hat{f}_{i,i-1/2}^D$ and $\hat{f}_{i,i+1/2}^D$ are the values of $\hat{f}_i^D$ in the faces of the $i-th$ control volume, $f_{i-1/2}$ and $f_{i+1/2}$ are the left and right fluxes post-correction by an approximate Riemann solver, g_L and g_R are the left and right correction functions, respectively. In Figure 3, the flow reconstruction process is represented.

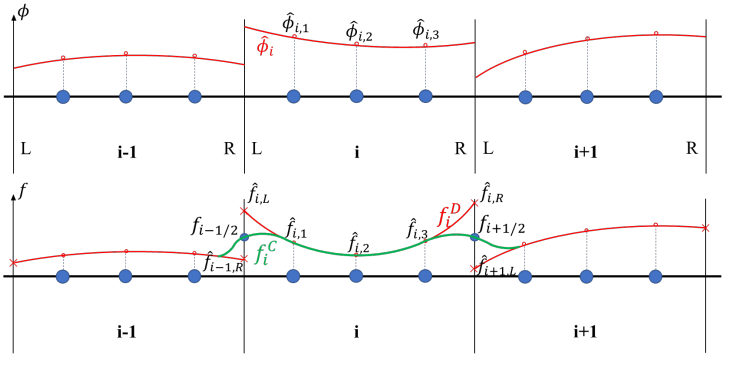


Figure 3: Reconstruction of the variable $\hat{\phi}^D$ and its continuous flux f^C.

The generic advective term $\frac{\partial \hat{f}_i^D}{\partial s}$ is approximated by:

$$\frac{\partial \hat{f}_i^D}{\partial s} = \sum_{j=1}^{P+1} \left[\hat{f}_{i,j}^D \frac{dL_j}{ds} + \left(f_{i-\frac{1}{2}} - \hat{f}_{i,i-\frac{1}{2}} \right) \frac{dg_L}{ds} + \left(f_{i+\frac{1}{2}} - \hat{f}_{i,i+\frac{1}{2}} \right) \frac{dg_R}{ds} \right] \tag{27}$$

Finally, the Eq. 22 can be rewritten as:

$$\frac{\partial \phi_i^D}{\partial t} = -\frac{1}{J_i} \sum_{j=1}^{P+1} \left[\hat{f}_{i,j}^D \frac{dL_j}{ds} + \left(f_{i-\frac{1}{2}} - \hat{f}_{i,i-\frac{1}{2}} \right) \frac{dg_L}{ds} + \left(f_{i+\frac{1}{2}} - \hat{f}_{i,i+\frac{1}{2}} \right) \frac{dg_R}{ds} \right] \tag{28}$$

where the term $(\partial \phi_i^D)/\partial t$ is the partial temporal derivative of the variable ϕ, for discontinuous ϕ on the faces of the $i-th$ control volume. In the present work, Eq. 28, in the context of the continuity equations, is expressed as:

$$\begin{aligned} \frac{\partial (\alpha_k \rho_k)_i^D}{\partial t} = & -\frac{2}{(x_{i+1} - x_i)} \sum_{j=1}^{P+1} \left[\widehat{(\alpha_k \rho_k u_k)}_{i,j}^D \frac{dL_j}{ds} + \left(\left(\alpha_k \rho_k u_k \right)_{i-\frac{1}{2}} - \right. \right. \\ & \left. \left. - \widehat{(\alpha_k \rho_k u_k)}_{i,i-\frac{1}{2}}^D \right) \frac{dg_L}{ds} + \left(\left(\alpha_k \rho_k u_k \right)_{i+\frac{1}{2}} - \widehat{(\alpha_k \rho_k u_k)}_{i,i+\frac{1}{2}}^D \right) \frac{dg_R}{ds} \right] \end{aligned} \tag{29}$$

And for the momentum equations, Eq. 28 is expressed as:

$$\begin{aligned} \frac{\partial u_{k_i}^D}{\partial t} = & -\frac{2u_k}{(x_{i+1} - x_i)} \sum_{j=1}^{P+1} \left[(\widehat{u})_{k_{i,j}}^D \frac{dL_j}{ds} + \left(u_{i-\frac{1}{2}} - \widehat{u}_{k_{i,i-\frac{1}{2}}} \right) \frac{dg_L}{ds} + \right. \\ & \left. + \left(u_{i+\frac{1}{2}} - \widehat{u}_{k_{i,i+\frac{1}{2}}} \right) \frac{dg_R}{ds} \right] + \frac{1}{\alpha_k \rho_k} RHS \end{aligned} \tag{30}$$

where the term RHS is the same as in Eq. 17. For the location of solution and flux points, we are using the Gauss-Lobatto quadrature, which ensures that the solution points are positioned on the control volume faces, thereby facilitating the implementation of the method. For the correction functions (g), two approaches were employed. The first uses Radau functions, leading to a simplified version of the NDG method (Hesthaven and Warburton (2007)), as follows:

$$g_{L,NDG} = \frac{1}{2} \left(\mathcal{L}_{P+1} - \mathcal{L} \right) \tag{31}$$

$$g_{R,NDG} = \frac{(-1)^{P+1}}{2} \left(\mathcal{L}_{P+1} - \mathcal{L}_P \right) \tag{32}$$

where $\mathcal{L}_P$ the Legendre polynomial of degree P. The second correction function used results in simplified version of the SD method (Huynh (2007),Galindez-Ramirez et al. (2022), Roe (1986)):

$$g_{L,SD} = \frac{1}{2} \mathcal{L}_{P+1}(1+s) \tag{33}$$

$$g_{R,SD} = \frac{(-1)^{P+1}}{2} \mathcal{L}_{P+1}(1-s) \tag{34}$$

where s is the parameterized space, Eq. 21. For the Riemann solver, we employed the Murman-Roe flux (Murman (1974), Morin, Flåtten and Munkejord (2013)), which can be written as follows:

$$f_{Roe} = \frac{1}{2}\left[f(\phi_R) + f(\phi_L) - |\hat{b}|(\phi_R - \phi_L)\right] \tag{35}$$

where ϕ_R and ϕ_L are the ϕ values on right and left of the face that flux f is evaluated, respectively. The ϕ advection speed at the interface $\hat{b}$ is given by:

$$\hat{b} = \begin{cases} \dfrac{f(\phi_R) - f(\phi_L)}{\phi_R - \phi_L}, & \text{if } \phi_R \neq \phi_L, \\ \dfrac{\partial f}{\partial \phi}, & \text{if } \phi_R = \phi_L. \end{cases} \tag{36}$$

3.3. Limiting Strategy

As a very high-order method, one of the challenges of using the FR approach is ensuring monotonicity. In this study, for that purpose, we employ the hierarchical Multidimensional Limiting Strategy (hMLP) (Park and Kim (2016)), which has previously been tested by Galindez-Ramirez et al. (2020a) and Galindo et al. (2023) in the context of 1-D FR, demonstrating excellent performance.

The hMLP procedure consists in rewriting ϕ^{Pn} as:

$$\phi_i^{Pn}(x) = \hat{\phi}_i + P1_i(x) + P2_i(x) + ... + Pn_i(x) \tag{37}$$

where Pn indicates the polynomial degree of this approximation and each $Pm(x)$ mode is defined as:

$$Pm_i(x) = \prod_m \phi_i^{Pn}(x) - \prod_{m-1} \phi_i^{Pn}(x), \quad 1 \leq m \leq Pn \tag{38}$$

where the projected solution $\prod_m \phi_i^{Pn}$ is obtained through the Legendre Vandermonde matrix (ν) (Hesthaven and Warburton (2007)), by computing the modal coefficients as:

$$\tilde{\phi}_i^{Pn} = \nu^{-1}\phi_i^{Pn} \tag{39}$$

Then, by truncating the modal coefficients, the projected solution of any m order can be obtained using the following equation:

$$\prod_m \phi_i^{Pn} = \nu \mathcal{J}\left[\tilde{\phi}_i^{Pn}\right], 1 \leq m \leq Pn \tag{40}$$

where $\mathcal{J}$ represents the truncation operator (Galindo et al. (2023), Hesthaven and Warburton (2007)).

3.4. Time Integration

Whenever solving time-dependent PDEs using high-order methods, it is crucial to carefully consider the selection of efficient time-stepping techniques. For instance, time integration can be performed using explicit methods such as the first-order Forward Euler scheme or total variation diminishing Runge-Kutta (TVD-RK) algorithms [66]. Since we are using very high-order spatial schemes, we opted for the third-order TVD-RK method. This choice provides an explicit method that is straightforward to implement, conditionally stable, and features a time step (Δt) dynamically adjusted according to the CFL (Courant-Friedrichs-Lewy) condition. The third-order TVD-RK method has already been used together with the FR method in other contexts (Galindez-Ramirez et al. (2020a), Galindez-Ramirez et al. (2022), Galindo et al. (2023)); however, this is the first application in the context of the two-fluid model. The model consists of the following steps:

$$\phi^1 = \phi^n + \Delta t[-R(\phi^n)] \tag{41}$$

$$\phi^2 = \frac{3}{4}\phi^n + \frac{1}{4}\left[\phi^1 + \Delta t\left(-R(\phi^1)\right)\right] \tag{42}$$

$$\phi^{n+1} = \frac{1}{3}\phi^n + \frac{2}{3}\left[\phi^2 + \Delta t\left(-R(\phi^2)\right)\right] \tag{43}$$

where $R(\phi^n)$ denotes the discrete approximation of $\frac{\partial f(\phi)}{\partial x}$, together with the RHS, both evaluated for ϕ at time step n or at an intermediate stage of the RK procedure. For classical schemes like the FOU and MUSCL schemes, the time step Δt is computed using the classical CFL calculation:

$$CFL = \frac{\Delta t \max|a|}{\Delta x} \tag{44}$$

where $max|a|$ is the maximum global wave velocity. For the FR method, the CFL condition is calculated considering the order of the approximation $\mathcal{O}$:

$$CFL = \frac{\Delta t \max|a|}{(\mathcal{O}+1)\Delta x} \tag{45}$$

3.5. SIMPLE Method

As done in other works that employ the two-fluid model (Morales-Ruiz et al. (2012),Wang et al. (2015)), a semi-implicit pressure-based method SIMPLE (Patankar (1980)) was used to couple the momentum, Eq. 2, and continuity equations, Eq. 1. As demonstrated by Morales-Ruiz et al. (2012) and Wang et al. (2015), this pressure-based segregated approach is robust, avoids the explicit calculation of the full Jacobian matrix, and exhibits a computational cost that is competitive with fully coupled methods, while maintaining good stability and performance in transient simulations of two-phase pipeline flows. Furthermore, separating the equations and solving the system through successive corrections simplifies the computational implementation and the use of high- and very-high-order methods.

A pressure-correction equation was used to update the pressure field and was obtained from the sum of the mass conservation equations for liquid and gas, dividing each equation, Eq. 18, by a representative value of the corresponding density ($\rho_{k_{ref}}$) (Wang et al. (2015), Morales-Ruiz et al. (2012)), and linearizing some terms to simplify the resolution (Morales-Ruiz et al. (2012)):

$$\frac{1}{\rho_{1_{ref}}}\left[(\alpha\rho)^k_{1,e}u^{k+1}_{1_e} - (\alpha\rho)^k_{1,W}u^{k+1}_{1_W}\right] + \frac{1}{\rho_{2_{ref}}}\left[(\alpha\rho)^k_{2,e}u^{k+1}_{2_e} - (\alpha\rho)^k_{2,W}u^{k+1}_{2_W}\right]$$

$$+\frac{\Delta x}{\Delta t}\left[\frac{(\alpha^k_1\rho^{k+1}_1)_P - (\alpha_1\rho_1)^n_P}{\rho_{1_{ref}}} + \frac{(\alpha^k_2\rho^{k+1}_2)_P - (\alpha_2\rho_2)^n_P}{\rho_{2_{ref}}}\right] = 0 \tag{46}$$

where the superscript k refers to the correction process after the pressure increment. Terms at k represent those evaluated at time $n+1$ but not yet updated by the pressure increment within the SIMPLE loop. Conversely, terms at $k+1$ correspond to those being updated by the pressure increment (δp):

$$p^{k+1} = p^k + \delta p \tag{47}$$

$$u^{k+1}_k = u^k_k + \left(\frac{A_e}{a_e}\right)_k\left(\delta p_P - \delta p_E\right) \tag{48}$$

where A_e and a_e are the coefficients of pressure term and of velocity u_e, respectively, in discretized momentum equation (Patankar (1980)). When the SIMPLE algorithm converges, the terms at levels $k+1$ and k become sufficiently

Algorithm 1: Two-Fluid Model with SIMPLE Scheme

1. Data input (geometry, initial and boundary conditions, total simulated time, tolerances)
2. Mesh generation (primal and dual mesh)
3. Initialization of variable fields $\left(u_1^{n+1}, u_2^{n+1}, \alpha_1^{n+1}, \alpha_2^{n+1}, \rho_1^{n+1}, \rho_2^{n+1}, p^{n+1}\right)$
4. Initialization of the residuals of velocities and pressure (e_u, e_p)
5. $t \leftarrow 0$ *(time index)*
6. **while** $t < t_{\text{total}}$ **do**
   - 6.1 time step calculation (Δt)
   - 6.2 $c \leftarrow 0$ *(SIMPLE correction counter)*
   - 6.3 **while** $(e_u >$ tol **or** $e_p >$ tol$)$ **and** $c < 10^3$ **do**
     - 6.3.1 Solve explicitly the momentum equations for velocities, Eqs. (16)–(17), using the previous pressure (p^k)
     - 6.3.2 Solve implicitly the pressure-correction equation, Eq. (46), using the new velocity fields
     - 6.3.3 Update explicitly pressure, Eq. (47), velocity, Eq. (48), and density fields, Eq. (8)
     - 6.3.4 Solve explicitly continuity equations for volume fractions, Eq. (18)
     - 6.3.5 Computation of residuals (e_u, e_p)
     - 6.3.6 Update $c \leftarrow c + 1$
   - 6.4 **end while**
   - 6.5 Temporal update: $u_1^n \leftarrow u_1^{n+1}$, $u_2^n \leftarrow u_2^{n+1}$, $\alpha_1^n \leftarrow \alpha_1^{n+1}$, $\alpha_2^n \leftarrow \alpha_2^{n+1}$, $\rho_1^n \leftarrow \rho_1^{n+1}$, $\rho_2^n \leftarrow \rho_2^{n+1}$, $p^n \leftarrow p^{n+1}$
   - 6.6 Update $t \leftarrow t + \Delta t$
7. **end while**
8. Post-processing

close, for this purpose, the residuals of the pressure (e_p) and velocities fields (e_u) are tested and considered converged to a tolerance (tol) of 10^{-3}:

$$e_p = \max\left(\frac{p_i^{k+1} - p_i^k}{p_i^k}\right) \tag{49}$$

The density increment term ρ_k^{k+1} is approximated using the Eq. 8. This scheme of successive approximations is also what allows the discretization of the advective terms of Eqs. 16 and 18 at time $n+1$ with an explicit method, since they can be treated at level k. The volume fractions are calculated only after the velocity and pressure fields have been updated. Consequently, the volume fractions are determined directly for the level $k+1$. The pressure equation is solved through the Two-Point Flux Approximation (TPFA) method, a.k.a Control Volume Finite Difference Method (CVFD), using the staggered grid arrangement. The global algorithm is presented in Algorithm 1.

4. Examples

In this section, we show the accuracy and efficiency of our numerical approach using five benchmark problems. To evaluate the error in our solution, we employ the L_1-norm, defined below:

$$E_{L_1} = \sum_{i=0}^{N-1} \Delta x_i |y_i - y_i^{ref}| \tag{50}$$

where y can be any parameter such as the pressure, the velocity of phase 1 or the volume fraction of phase 2. The convergence rate, using the L_1-norm of the errors, on two consecutive grids (x_1) and (x_2) was computed by:

$$R_{L_1} = \frac{\log[E_{L_1}(x_1)] - \log[E_{L_2}(x_2)]}{\log(\Delta x_1) - \log(\Delta x_2)} \tag{51}$$

Another metric used is the relative error (E_{r_i}), which is calculated with respect to a reference result (y_i^{ref}):

$$E_{r_i} = \left| \frac{y_i - y_i^{ref}}{y_i^{ref}} \right| \tag{52}$$

4.1. 1-D Transport problem with a non-linear flux and a smooth solution

This case is a simple non-linear model essentially inviscid Burgers' problem (Galindo et al. (2023)) in the smooth stage of the solution, that can be interpreted as a particular case of Eq. 2. Considering these hypotheses, the Burgers' equation is written, in conservative form, as:

$$\frac{\partial u}{\partial t} + \frac{\partial F(u)}{\partial x} = 0 \tag{53}$$

in which:

$$F(u) = \frac{u^2}{2} \tag{54}$$

Periodic boundary conditions are applied at the domain boundaries:

$$u(x = 0, t) = u(x = L, t), \ \forall\, t \in T \tag{55}$$

where L is the length of the spatial domain and $T = [0.0.1]$. The initial condition is given by a Gaussian curve:

$$u(x, t = 0) = 0.5 e^{-\frac{(x-0.5)^2}{0.01}}, \ \ \forall x \in \Omega \tag{56}$$

where $\Omega = [0.1]$.

We evaluate the convergence rates of the FR using the correction functions that lead to particular versions of NDG and SD methods. For comparison purposes, we have also included the results of the FOU and the MUSCL methods. We compute the L_1-norm (E_{L_1}) of the errors and the numerical convergence rates (R_{L_1}), for the smooth solution at $t = 0.1$. The results are presented in Table 1 and Figure 4. The simulations are made using a sequence of uniformly refined meshes from 32 to 512 CV, similar as done in Galindez-Ramirez et al. (2020a). For all meshes, we have adopted a $CFL = 0.9$ for the FR and FOU schemes, and $CFL = 0.45$ for the MUSCL scheme, values close to the stability limits of these methods for explicit schemes (van Leer (1979), Romero et al. (2016), Toro (2009), LeVeque (2002)).

For all schemes tested, the convergence behavior is close to the theoretical asymptotic order of accuracy, except for the fourth-order FRs (FR-P3), which seems to show a small decrease in the convergence rate in relation to the asymptotic order, as also observed in Galindo et al. (2023). For this problem, increasing the order of the approximation in the FR showed significant gains in accuracy. From Table 1, it is clear that the classical FOU method, normally used in commercial oil pipeline 1-D flow simulators, is less accurate than all other methods and for all presented meshes. In fact, for this particular problem, only for the mesh with 512 CV, the FOU method presents a similar accuracy to the less accurate version of the high order methods, i.e., to the MUSCL and to the FR-SD-P1 and FR-NDG-P1 methods with 64 CV. In Figure 4, we also present errors and computational time for all methods tested. From this figure, it is clear that the MUSCL method is slightly faster than both, the FR-NDG-P1 and FR-SD-P1 methods that formally have the same order of approximation. However, in terms of accuracy the FR-NDG-P1 outperforms the MUSCL scheme, which is slightly more accurate than the FR-SD-P1 for all the meshes used. Besides Considering the P2 and P3 polynomials reconstructions, for all meshes resolutions used and for the same polynomial order, the FR-NDG version is more

Table 1
Error and convergence rates for the L_1-norm of error of the inviscid Burgers' problem.

CV	32		64		128		256		512	
Method	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}
FOU	7.83E-03	-	4.36E-03	0.85	2.31E-03	0.91	1.21E-03	0.94	6.14E-04	0.97
MUSCL	3.33E-03	-	1.26E-03	1.41	3.82E-04	1.72	1.04E-04	1.88	2.73E-05	1.93
FR-NDG-P1	1.99E-03	-	5.81E-04	1.78	1.58E-04	1.87	4.18E-05	1.92	1.08E-05	1.96
FR-SD-P1	2.74E-03	-	8.62E-04	1.67	2.40E-04	1.85	6.41E-05	1.90	1.66E-05	1.95
FR-NDG-P2	2.66E-04	-	3.39E-05	2.97	4.38E-06	2.95	5.57E-07	2.98	7.01E-08	2.99
FR-SD-P2	3.57E-04	-	5.09E-05	2.81	6.93E-06	2.88	9.01E-07	2.94	1.15E-07	2.97
FR-NDG-P3	2.69E-05	-	2.08E-06	3.70	1.47E-07	3.82	1.07E-08	3.78	8.78E-10	3.61
FR-SD-P3	4.10E-05	-	3.12E-06	3.72	2.23E-07	3.81	1.58E-08	3.82	1.19E-09	3.73

accurate than the FR-SD variant for this particular non-linear problem. Besides, from Figure 4, it is also clear that the FR-NDG-P3 and the FR-SD-P3 have a better cost-benefit relation than the FOU and MUSCL methods and the other FR polynomial approximations (i.e., FR-NDG-P1 and -P2, and FR-SD-P1 and P2). Finally, we can also see that the FR-NDG behaves slightly better than the FR-SD variant.

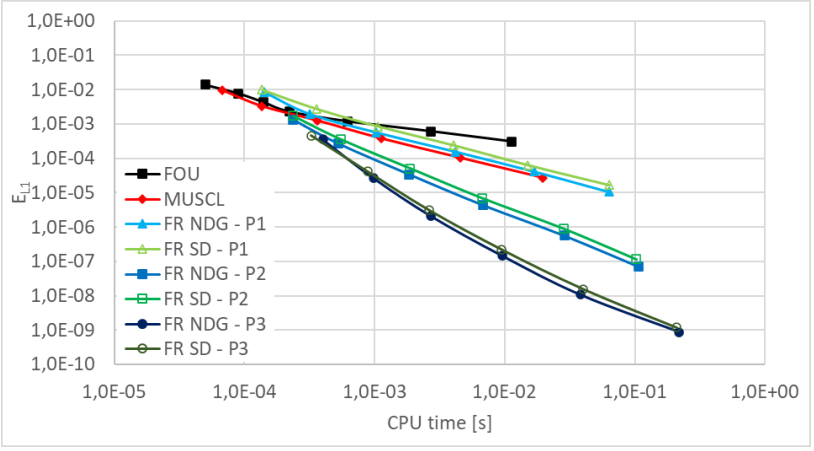


Figure 4: CPU times and norm L1-errors for the 1-D problem with a non-linear flux and a smooth solution.

4.2. Discontinuity Movement in the Uniform Flow Case

In this classical example, adapted from Yeom and Chang (Yeom and Chang (2011)), a discontinuity moves through a two-phase flow of a liquid and a gas in a 12-meter-long horizontal pipe under constant pressure and velocities. In this case, the liquid and the gas phases have the same velocity and there is no formation of $K - H$ instabilities. Besides, there is a diaphragm in the midsection of the horizontal pipe, resulting in different states on the left and right sides of the diaphragm (Morales-Ruiz et al. (2012)). In Figure 5, we present a schematic of the flow in the 1-D, the diaphragm with boundary conditions, the initial state and the state after 0.1 second. The velocities are $u_L = u_G = 10\ m/s$ and the pressure is $p = 10^6\ \ Pa$ throughout the entire domain. Initially, to the left of the midsection, the liquid volume fraction is $\alpha_L = 0.8$, while to the right it is $\alpha_L = 0.2$.

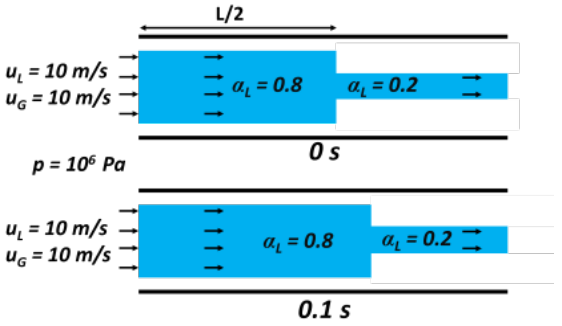


Figure 5: Two-Phase Flow with a Moving Discontinuity: the initial state and the state after 0.1 second.

The pipe is assumed to be adiabatic, and frictional forces are neglected. The gas void fraction (α_G) is advected along the computational domain when the flow begins. For this case, we used $CFL = 0.9$ for the FR and FOU schemes, and $CFL = 0.45$ for the MUSCL scheme. The solutions are obtained using 32, 64, 128, 256 and 512 CV for the instant of time $t = 0.1\ s$. To evaluate the performance of our formulation, in Figure 6, we present the air void fraction (α_G) for different mesh resolutions with 32, 64 and 128 CV and different methods, FOU, MUSCL, FR-NDG-P1, P2 and P3 and for FR-SD-P1, P2 and P3.

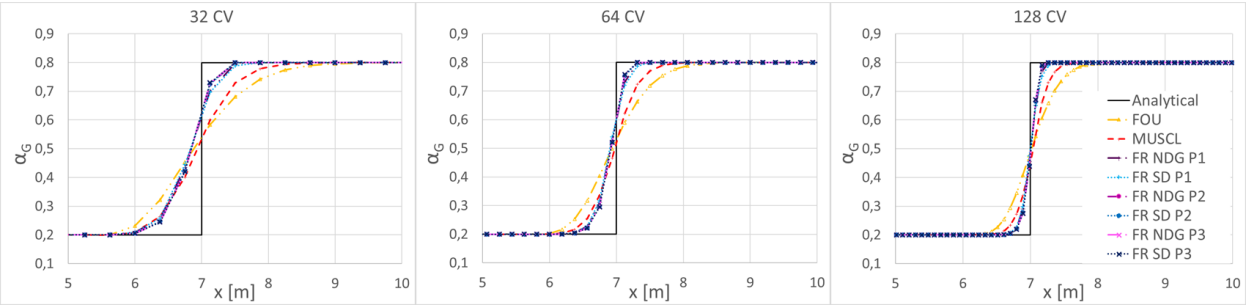


Figure 6: Two-Phase Flow with a Moving Discontinuity: Air void fraction (α_G), $t = 0.1s$, for different grid resolutions (32, 64 and 128 CV) and different methods (FOU, MUSCL, FR-NDG-P1, FR-NDG-P1, P2, P3 and for FR-SD-P1, P2 and P3).

For this case, with a moving discontinuity, the MUSCL and all variants of the FR method presented more accurate results than the FOU method. Besides, due to the discontinuity of the void fraction it is well known that the theoretical convergence rates cannot be achieved (LeVeque (2002), Harten (1983)). In Table 2 we present the L_1-norm of the errors, and the convergence rates the FOU, MUSCL and all variants of the FR method we have presented in the present work. Besides, even the lowest orders of the FR method, i.e., the FR-SD-P1 and the FR-NDG-P1 were more accurate than the MUSCL scheme, possibly because FR performs interpolation using the reconstruction polynomials, whereas MUSCL relies on extrapolation. It is worth mentioning that except for the case with P1 reconstruction where the FR-NDG-P1 performed slightly better than the FR-SD-P1 for all grid resolutions, the FR-NDG and the FR-SD presented essentially the same accuracy for different grid resolutions.

4.3. Water Faucet Problem

In this example, we solve the classical water faucet problem (Harten (1983)). In the present paper, we consider a vertical pipe with 12 meters length, filled with water and air where water is injected at the top of the duct and accelerates downward due to gravity and due to the pressure gradient, air influx also occurs at the bottom of the pipe (Wang et al. (2015), Morales-Ruiz et al. (2012)). In this case, we also assume that frictional forces effects are negligible. Initially, the void fraction is $\alpha_G = 0.2$ throughout the whole duct and the water volume fraction $\alpha_W = 0.8$, as suggested in Figure 7. A discontinuity in the volumetric fractions of water and gas occur when the water falls downward under the force of gravity with a velocity of $u_W = 10\ m/s$ from the top of the duct.

Table 2
L_1-norm of errors for discontinuity moving case.

CV	32		64		128		256		512	
Method	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}
FOU	3.15E-01	-	2.27E-01	0.47	1.57E-01	0.53	1.12E-01	0.49	7.86E-02	0.51
MUSCL	2.16E-01	-	1.58E-01	0.45	9.46E-02	0.74	6.52E-02	0.54	4.12E-02	0.66
FR-NDG-P1	1.52E-01	-	9.16E-02	0.73	5.25E-02	0.80	3.29E-02	0.68	2.09E-02	0.66
FR-SD-P1	1.61E-01	-	9.48E-02	0.76	5.82E-02	0.71	3.54E-02	0.71	2.33E-02	0.60
FR-NDG-P2	1.31E-01	-	8.45E-02	0.64	4.67E-02	0.86	2.94E-02	0.67	1.79E-02	0.72
FR-SD-P2	1.31E-01	-	8.45E-02	0.64	4.67E-02	0.86	2.94E-02	0.67	1.79E-02	0.72
FR-NDG-P3	1.28E-01	-	8.29E-02	0.62	4.49E-02	0.88	2.81E-02	0.67	1.67E-02	0.76
FR-SD-P3	1.28E-01	-	8.29E-02	0.62	4.49E-02	0.88	2.81E-02	0.67	1.67E-02	0.76

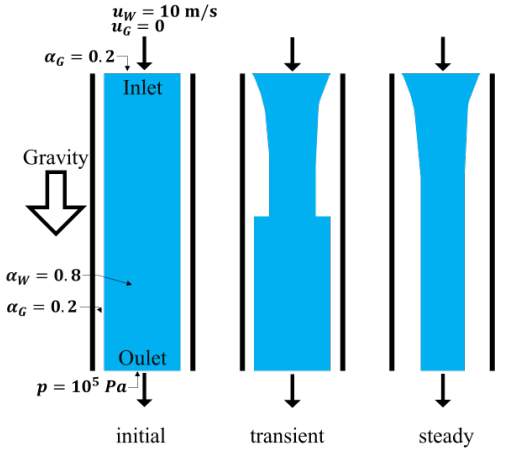


Figure 7: Water faucet problem: Initial configuration (left), transient profile (center) and steady state solution (right) (Adapted from Morales-Ruiz et al. (2012)).

Just like in the previous examples, the solutions are obtained using 32, 64, 128, 256 and 512 CV, in Figure 8, a zoom of the results of the air void fraction (α_G) is shown for time $t = 0.5$ s and part of the meshes. For this case, we used $CFL = 0.5$ for the FR and FOU schemes, and $CFL = 0.25$ for the MUSCL scheme, because this example is more unstable than the previous ones, due to the existence of relative speed between the phases.

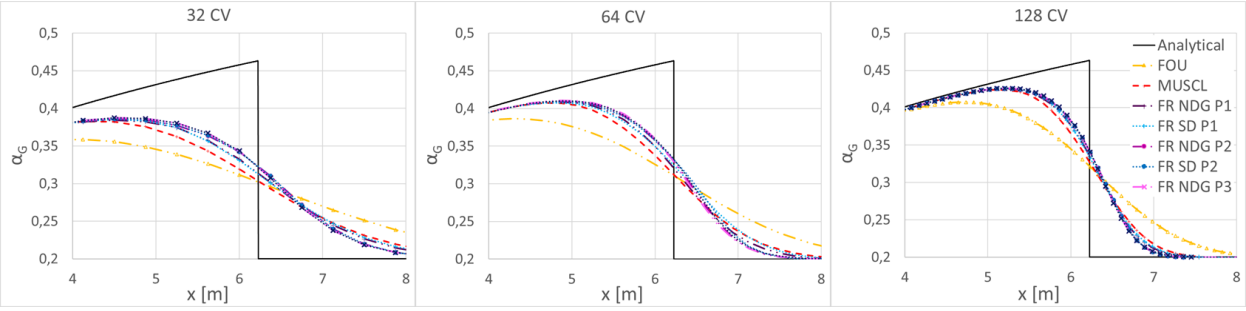


Figure 8: Air void fraction (α_G), $t = 0.5$ s, for different grid resolutions (32, 64 and 128 CV) and different methods (FOU, MUSCL, FR-NDG-P1, FR-NDG-P1, P2, P3 and for FR-SD-P1, P2 and P3).

The FR schemes demonstrate superior accuracy in capturing the shock region and, consistent with the previous case, produce results that are closely aligned. In Table 3, we compute the L_1-norm of the errors and the convergence rates of the results of this case.

Table 3
L_1-norm of errors for water faucet problem.

CV	32		64		128		256		512	
Method	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}	E_{L_1}	R_{L_1}
FOU	4.62E-01	-	3.05E-01	0.60	2.05E-01	0.57	1.40E-01	0.55	9.58E-02	0.55
MUSCL	3.54E-01	-	2.20E-01	0.68	1.39E-01	0.66	8.89E-02	0.64	5.95E-02	0.58
FR-NDG-P1	3.36E-01	-	2.06E-01	0.71	1.27E-01	0.70	8.22E-02	0.63	5.69E-02	0.53
FR-SD-P1	3.39E-01	-	2.07E-01	0.71	1.28E-01	0.69	8.27E-02	0.63	5.72E-02	0.53
FR-NDG-P2	3.23E-01	-	1.98E-01	0.71	1.24E-01	0.67	8.11E-02	0.61	5.28E-02	0.62
FR-SD-P2	3.23E-01	-	1.97E-01	0.71	1.24E-01	0.67	8.11E-02	0.61	5.28E-02	0.62
FR-NDG-P3	3.16E-01	-	1.94E-01	0.71	1.23E-01	0.66	8.09E-02	0.60	5.78E-02	0.48
FR-SD-P3	3.16E-01	-	1.94E-01	0.71	1.23E-01	0.66	8.09E-02	0.60	5.78E-02	0.48

Again, convergence rates do not reach nominal values due to the presence of the shock. Furthermore, it is also observed that for the finest mesh (512 CV), the results of the FR-P3 schemes are greater than the error of FR-P2. Most likely, this is attributed to the slope-limiting process typical of all high-order methods, necessary to ensure stability at the shock front. This process can contaminate the solutions in the smooth region behind the shock, deteriorating the optimal order of accuracy (Galindez-Ramirez et al. (2020a)).

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4.3.1. Hyperbolicity analysis

The water faucet problem represents a particularly demanding configuration for the numerical solution of the two-fluid four-equation model. In this case, the flow involves relative motion between the phases, strong acceleration due to gravity, and the development of sharp gradients in the volume fraction field. These characteristics increase the susceptibility of the model to numerical instabilities and may lead to local loss of hyperbolicity.

To further investigate the numerical behavior of the proposed formulation, an additional analysis was performed focusing on the stability and robustness of the FR method under these challenging conditions.

For the water faucet problem, the governing equations can be written in quasi-linear form as

$$\frac{\partial \mathbf{U}}{\partial t} + \mathbf{G}\frac{\partial \mathbf{U}}{\partial x} = \mathbf{D} \tag{57}$$

$$\Delta = \frac{\alpha_G \rho_L u_G}{c_G^2} + \frac{\alpha_L \rho_G u_L}{c_L^2} \tag{58}$$

$$\mathbf{G} = \begin{bmatrix} \frac{1}{\Delta}\left(\frac{\alpha_L\rho_G u_G}{c_L^2} + \frac{\alpha_G\rho_L u_L}{c_G^2}\right) & \frac{\alpha_L\alpha_G\rho_G(u_G-u_L)}{c_L^2 c_G^2\Delta} & \frac{\alpha_L\alpha_G\rho_G}{c_L^2\Delta} & -\frac{\alpha_G\alpha_L\rho_L}{c_G^2\Delta} \\ \frac{\rho_L\rho_G(u_G-u_L)}{\Delta} & \frac{1}{\Delta}\left(\frac{\alpha_L\rho_G}{c_L^2} + \frac{\alpha_G\rho_L}{c_G^2}\right) & \frac{\rho_L\alpha_G\rho_G}{\Delta} & \frac{\rho_G\alpha_L\rho_L}{\Delta} \\ \frac{\Delta p_{art}}{\alpha_G\rho_G} & \frac{1}{\rho_G} & u_G & 0 \\ -\frac{\Delta p_{art}}{\alpha_L\rho_L} & \frac{1}{\rho_L} & 0 & u_L \end{bmatrix} \tag{59}$$

The eigenvalues of matrix $\mathbf{G}$ are obtained from the characteristic polynomial

$$P(\lambda) = \det(\mathbf{G} - \lambda\mathbf{I}).$$

The appearance of complex eigenvalues indicates a local loss of hyperbolicity of the system. The procedure adopted to evaluate the eigenvalues along the computational domain is summarized in Algorithm 2.

Algorithm 2: Eigenvalue Computation for System Stability

1. Input: $\mathbf{u}$, α, ρ, c, Δp_{art}
2. Set $\epsilon \leftarrow 10^{-8}$
3. **for** $i = 1$ to N **do**
   - 3.1 Compute model terms
   - 3.2 $D \leftarrow \dfrac{\alpha_{G,i}\rho_{L,i}u_{G,i}}{c_G^2} + \dfrac{\alpha_{L,i}\rho_{G,i}u_{L,i}}{c_L^2} + \epsilon$
   - 3.3 Assemble matrix $\mathbf{G}$
   - 3.4 Compute eigenvalues: $\lambda_i \leftarrow \text{eigvals}(\mathbf{G})$
4. **end**
5. Export $\{i, \text{Re}(\lambda_i), \text{Im}(\lambda_i)\}$ to CSV

Figure 9 presents the volumetric fraction (α_G) distribution along the domain, comparing the analytical solution with the numerical results obtained using the MUSCL scheme. Red markers identify control volumes where the eigenvalues become complex, characterizing regions of local loss of hyperbolicity.

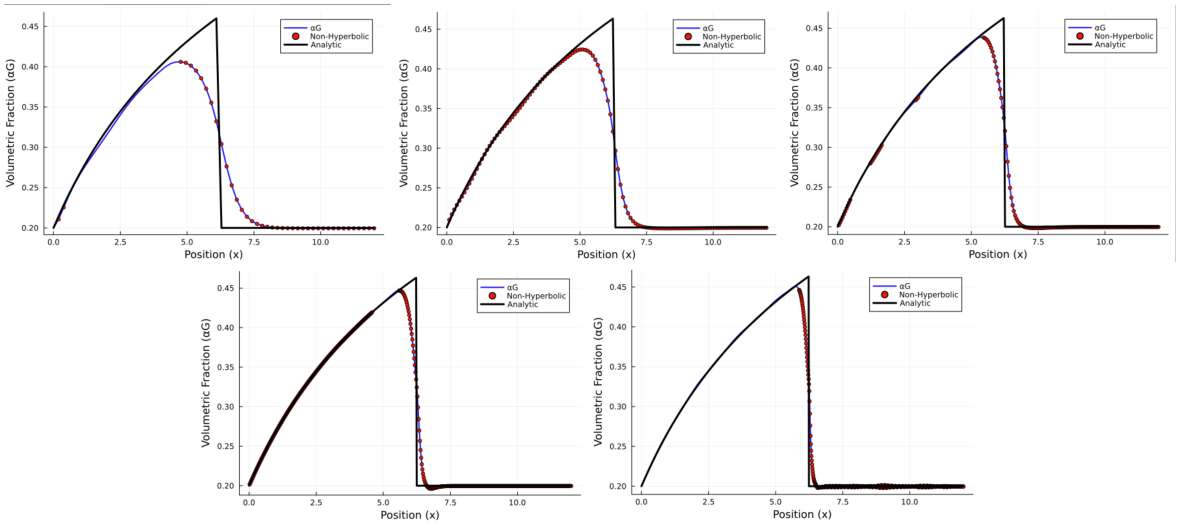


Figure 9: Local loss of hyperbolicity with MUSCL - 64,128, 256, 512, 1024 CV

Despite the presence of non-hyperbolic regions, the numerical solution remains stable and convergent. This behavior is associated with the intrinsic numerical diffusion of the MUSCL scheme and the action of slope limiters, which act as regularization mechanisms that suppress the growth of non-physical instabilities. As the mesh is refined, the regions with complex eigenvalues tend to concentrate around the high-gradient zones, while the numerical solution approaches the analytical profile.

Figure 10 shows the corresponding results obtained with the FR scheme. In comparison with the MUSCL results, the regions with complex eigenvalues exhibit a larger spatial extent, particularly in zones where the α_G gradient is still developing, before reaching the maximum slope of the discontinuity.

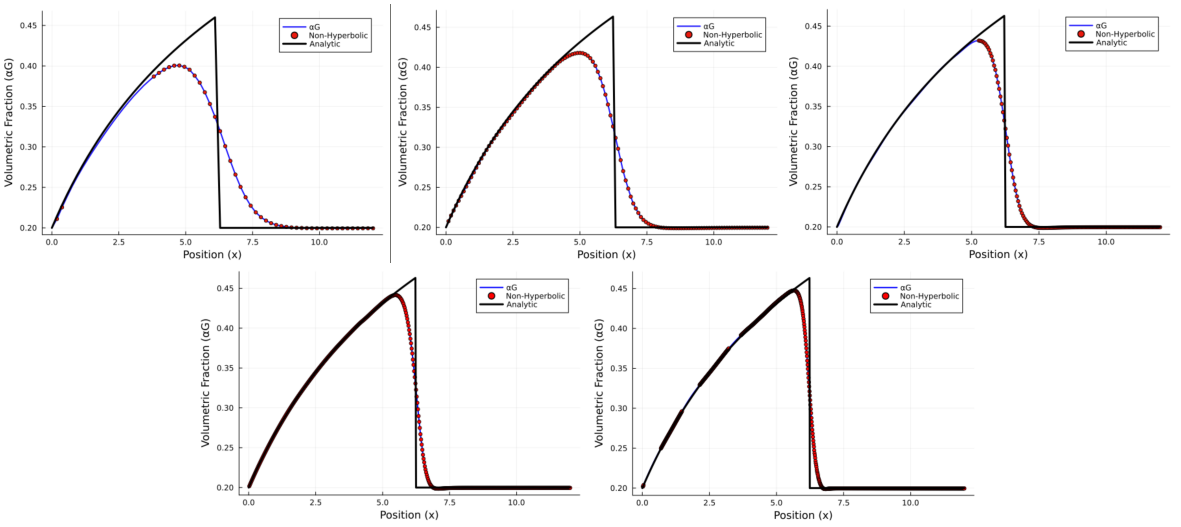


Figure 10: Local loss of hyperbolicity with FR - 64,128, 256, 512, 1024 CV

This behavior is attributed to the high-order polynomial reconstruction inherent to the FR method, which provides enhanced sensitivity to incipient gradients. As a consequence, the FR scheme is able to detect the onset of non-hyperbolic conditions earlier than lower-order methods.

From this perspective, the high-order FR formulation acts not only as an accurate discretization technique but also as a diagnostic tool for identifying the loss of hyperbolicity inherent to the two-fluid four-equation model. It is important to note that the two-fluid single-pressure formulation is known to become locally ill-posed when hyperbolicity is lost. Therefore, the ability of the proposed FR framework to produce stable and physically consistent solutions, even in the presence of complex eigenvalues, demonstrates the robustness of the numerical approach and the effectiveness of the adopted stabilization mechanisms.

4.4. Oil-Water Annular Flow

In this case, we consider the downward vertical annular flow of water and oil in a duct with a length of 100 meters and a diameter $D = 10\ cm$. The effects of shear at the fluid interface and along the pipe wall are considered as suggested in Figure 11. Both phases are injected at the top of the pipe with the same velocity of $u_w = u_o = 3.0$ m/s. Fluid densities and viscosities are $\rho_o = 823.8\ \mathrm{kg/m^3}$ and $\rho_w = 988.5\ \mathrm{kg/m^3}$, and viscosities of $\mu_o = 2.80\ \mathrm{mPa \cdot s}$ and $\mu_w = 0.55\ \mathrm{mPa \cdot s}$ for oil and water, respectively, at the inlet. Each phase is injected, occupying half of the cross-section ($\alpha_w = 0.5$), and the temperature throughout the pipe is maintained at 50 °C. At the outlet, the pressure is $p = 2.0$ MPa. Based on the boundary conditions, the Reynolds numbers are $Re_w \approx 3.8 \times 10^5$ and $Re_o \approx 6.2 \times 10^4$, which are consistent with traditional oil well values (Hasan and Kabir (2018), Brill and Mukherjee (1999)).

The objective of this case is to verify the accuracy of our high-order FR methods for the Two-Fluid Four-Equation formulation applied to a more complex and realistic problem considering an annular two-phase flow of oil-water in a vertical pipe. The interfacial friction term (τ_i) and the wall friction term (τ_w) are taken into account and evaluated according to Cazarez-Candia and Piedra-González (2017). To obtain a reference solution we use the commercial software ALFAsim (ESSS (2024)) which uses a more complex representation of the fluid flow based on a multifield formulation. Results are compared after the steady-state solution is achieved.

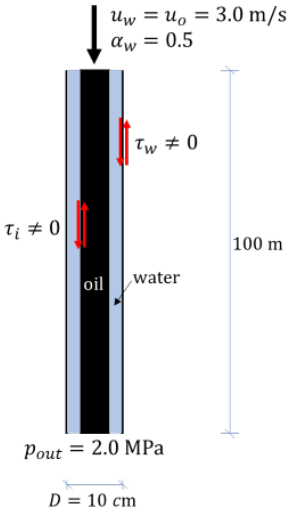


Figure 11: Oil-water annular flow case: geometry and boundary conditions.

Figure 12 (a,b,c) presents, respectively the results for the pressure field (p), water velocity (u_w) and the oil volumetric fraction (α_o) throughout the pipe, using both, the commercial software ALFAsim and our in-house software for some of our developed and implemented formulations. For this example, we restrict our attention to the FOU, MUSCL, and the FR-NDG-P1 and the FR-SD-P1 schemes. We must keep in mind that the physical model used by the ALFAsim software is more complex and presents certain differences when compared to the simpler Two-Fluid Four-Equation single-pressure model we have used in the present work and some differences are expected for the numerical solution using both formulations. For this problem, we have performed a grid convergence study, using the FOU scheme, and we have observed that our solution exhibits a relative error at the inlet of approximately $E_{r,inlet}(p) = 1.55\%$ compared to the pressure solution obtained by the ALFAsim with a grid resolution of 1024 CV and we have decided to use this grid for our subsequent analysis.

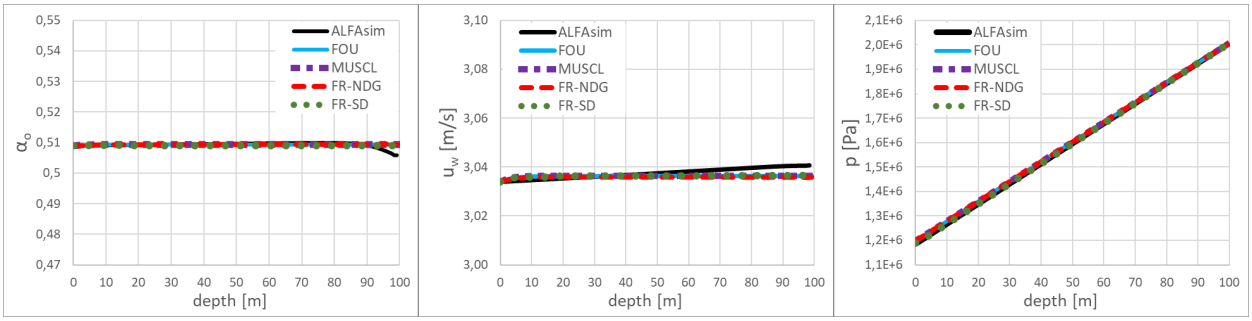


Figure 12: Viscous downward flow results, with 128 CV.

As previously mentioned, when compared to the commercial software ALFAsim, our formulation has produced a linear pressure field with a difference of only 1.55% on the pipe inlet and for the water velocity field and the oil volume fraction, the maximum relative errors were $E_{r,max}(u_w) = 0.2\%$ and $E_{r,max}(\alpha_o) = 0.7\%$, respectively for the grid with 128 CV. These errors do not vanish with further grid refinement and we attribute them to differences in empirical modeling of the friction terms at the phase interface and the pipe wall, which we believe were handled using a different model in ALFAsim.

4.5. Annular Upward Flow in a Long Pipe

In this case, the upward vertical annular flow of water and oil in a long pipe is considered. The pipe has a length of $L = 1000$ m and an internal diameter of $D = 0.10$ m. The effects of shear stress at the fluid interface and along the pipe wall are taken into account, as illustrated in Figure 13. This case is intended to represent the upward propagation

of a sudden increase in water cut in a petroleum well. Initially, water and oil occupy equal volumetric fractions along the entire well, i.e., $\alpha_w = 0.5$. Subsequently, a water front with $\alpha_w = 0.7$ is introduced at the bottom of the well and propagates upward.

Heavy oil conditions are assumed, with densities of $\rho_o = 985\ \mathrm{kg/m^3}$ and $\rho_w = 990\ \mathrm{kg/m^3}$, and dynamic viscosities of $\mu_o = 0.84\ \mathrm{Pa \cdot s}$ and $\mu_w = 0.55\ \mathrm{mPa \cdot s}$ for oil and water, respectively, at the inlet. To represent realistic production conditions (Speight (2019),Hasan and Kabir (2018),Brill and Mukherjee (1999)), a constant outlet pressure of $p = 30.0$ MPa is prescribed, and the temperature is assumed to be uniform along the pipe at $T = 80.0\ ^\circ\mathrm{C}$. The superficial velocities of both phases are set equal, $u_w = u_o = 3.0\ \mathrm{m/s}$, and the well is considered adiabatic.

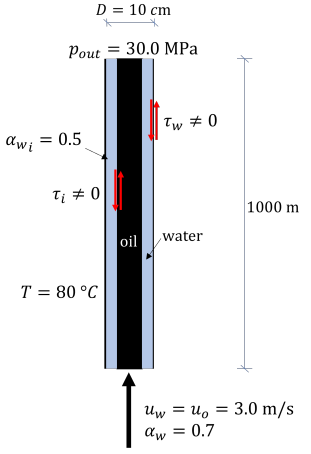


Figure 13: Annular Upward Flow in a Long Pipe Case: geometry and boundary conditions.

Figure 14 presents the results at $t = 150$ s for water volume fraction (α_w). In *ALFAsim*, two spatial discretizations were employed, with 128 CV and 7500 CV. In contrast, in the implemented in-house simulator, only the coarse mesh with 128 CV was used in combination with the Flux Reconstruction (FR) methods.

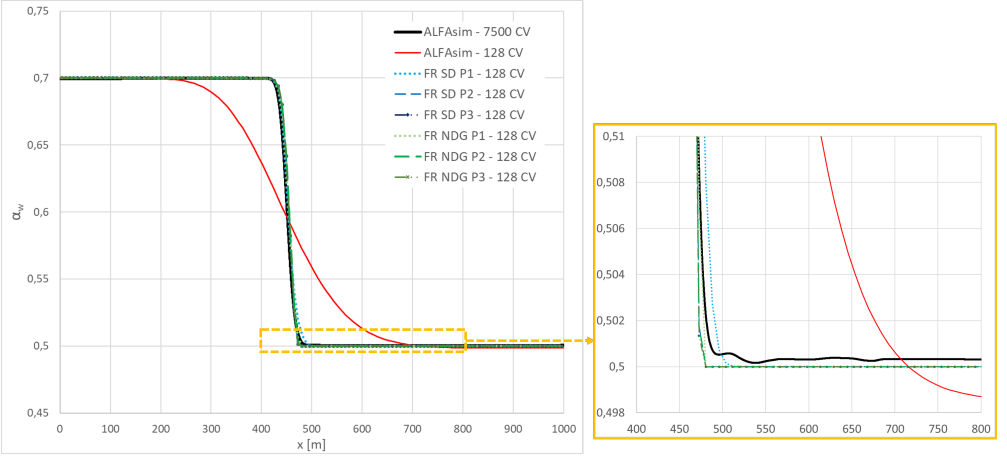


Figure 14: Annular Upward Flow in a Long Pipe Case: results of water volume fraction.

The results obtained with the FR method on the coarse mesh of 128 CV were equivalent to those obtained with *ALFAsim* using the refined mesh of 7500 CV. The NDG version of the FR scheme exhibited a slight improvement in accuracy in the shock resolution when compared to the SD version. For this case, the third- and fourth-order approximations (P2 and P3) produced virtually identical results.

These results indicate that the implemented in-house simulator is capable of delivering accurate predictions even for a highly complex flow scenario.

5. Conclusions

In the present work, for the first time in the literature we adapt and implement the very high-order Flux Reconstruction (FR) method to discretize the advective terms of the Two-Fluid Four-Equation (TFFE) single-pressure to model the two-phase flow in pipelines. In our implementation we considered the FR with two correction functions: one leading to a particular version of the Nodal Discontinuous Galerkin (NDG) method (FR-NDG), and the other resulting in a specific version of the Spectral Difference (SD) method (FR-SD). The method was used to discretize the advective terms of the momentum and mass conservation equations of the TFFE model and it was implemented with polynomial order polynomial approximation. In our formulation the diffusive terms of the model were discretized by the classical Control Volume Finite Difference (CVFD), and the time discretization was performed by a third-order Total Variation Diminishing Runge-Kutta (TVD-RK) scheme and to formally guarantee monotone solutions free of spurious oscillations, we have used a limiting procedure (hMLP). In order to evaluate the accuracy and robustness of our formulation, we have solved some benchmark problems found in the literature and compare the results with the classical First Order Upwind (FOU) method and the 2nd Order Monotone Upwind-centered Scheme for Conservation Laws (MUSCL) scheme, both using first order Euler Forward time integration. For all test cases the FR-NDG and the FR-SD (P1, P2 and P3) were superior in terms of accuracy than the FOU and the MUSCL methods and the FR-NDG was slightly superior to the FR-SD variant, besides the FR-NDG-P3 presented the better relation between accuracy and CPU time. We have also solved a more complex oil-water two-phase annular fluid flow and compared our results with the commercial software ALFAsim. For the cases analyzed, our results were very promising. In the near future we intend to model more complex flow problems, involving thermal effects and flow pattern prediction. Besides, we also intend to incorporate an automatic h-p adaptation strategy to improve the overall performance of our formulation.

CRediT authorship contribution statement

Anderson V. do Nascimento: Investigation, Writing - original draft, Methodology, Validation, Visualization, Software, Formal analysis, Conceptualization, Data curation. **Cleuber E. N. Silva:** Software, Validation, Visualization, Writing - original draft. **Giselle M. L. L. da Silva:** Funding acquisition, Supervision, Resources, Project administration. **Tiago C. Freitas:** Funding acquisition, Supervision, Resources. **Rosane A. Fontes:** Funding acquisition, Supervision, Resources. **Mônica T. da Silva:** Funding acquisition, Supervision, Resources. **Jussara de M. Silva:** Funding acquisition, Supervision, Resources, Project administration. **Paulo R. M. Lyra:** Supervision, Resources, Writing - review and editing, Funding acquisition, Formal analysis, Investigation. **Alessandro R. E. Antunes:** Supervision, Resources, Writing - review and editing, Project administration, Formal analysis, Investigation. **Darlan K. E. de Carvalho:** Writing - review and editing, Project administration, Supervision, Resources, Funding acquisition, Formal analysis, Investigation.

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